Limit

90. For $\alpha, \beta, \gamma \in \mathbf{R}$, If $\lim\limits_{x\to 0}\dfrac{x^2\sin\alpha x+(\gamma-1)e^{x^2}}{\sin 2x-\beta x}=3$, then $\beta+\gamma-\alpha$ is equal to :

यदि $\alpha, \beta, \gamma \in \mathbf{R}$, के लिए $\lim\limits_{x\to 0}\dfrac{x^2\sin\alpha x+(\gamma-1)e^{x^2}}{\sin 2x-\beta x}=3$ है, तो $\beta+\gamma-\alpha$ बराबर है :

(A) 7 (B) 6 (C) 4 (D) –1

Ans. A

Sol. $\lim\limits_{x\to 10}\dfrac{x^2(\alpha x)+(\gamma-1)\left(1+\dfrac{x^2}{1}\right)}{2x-\dfrac{8x^3}{6}-\beta x}=3$

$\lim\limits_{x\to 0}\dfrac{(\gamma-1)+(\gamma-1)x^2+\alpha x^3}{(2-\beta)x-\dfrac{4}{3}x^3}$

$\gamma-1,\beta=2,-\dfrac{3\alpha}{4}=+3 \qquad \Rightarrow \alpha=4$

$\beta+\gamma-\alpha=7$

91. If $\lim\limits_{x\to 0}\dfrac{\cos(2x)+a\cos(4x)-b}{x^4}$ is finite, then (a + b) is equal to :

यदि $\lim\limits_{x\to 0}\dfrac{\cos(2x)+a\cos(4x)-b}{x^4}$ परिमित है, तो (a + b) बराबर है :

(A) $\dfrac{3}{4}$ (B) 0 (C) $\dfrac{1}{2}$ (D) –1

Ans. C

Sol. $\lim\limits_{x\to 0}\dfrac{\cos(2x)+a\cos(4x)-b}{x^4}$

$\lim\limits_{x\to 0}\dfrac{\left(1-\dfrac{(2x)^2}{2!}+\dfrac{(2x)^4}{4!}+....\right)+a\left(1-\dfrac{(4x)^2}{2!}+\dfrac{(4x)^4}{4!}-.....\right)-b}{x^4}$

For limit to be finite
$1 + a - b = 0$
And $-2 - 8a = 0$
$\Rightarrow 8a = -2$

$\Rightarrow a = \frac{-1}{4}$

$\Rightarrow b = 1 + a$

$= 1 - \frac{1}{4}$

$b = -\frac{3}{4}$

$a + b = \frac{-1}{4} + \frac{3}{4} = \frac{1}{2}$

92. If $\lim\limits_{x \to 1^+} \frac{(x-1)(6+\lambda\cos(x-1)) + \mu\sin(1-x)}{(x-1)^3} = -1$, where $\lambda, \mu \in R$, then $\lambda + \mu$ is equal to :

यदि $\lim\limits_{x \to 1^+} \frac{(x-1)(6+\lambda\cos(x-1)) + \mu\sin(1-x)}{(x-1)^3} = -1$, $\lambda, \mu \in R$ हैं, तो $\lambda + \mu$ बराबर है :

(A) 19 (B) 17 (C) 18 (D) 20

Ans. C

Sol. $\lim\limits_{x \to 1^+} \frac{(x-1)\left(6+\lambda\cos(x-1)\right) + \mu\sin(1-x)}{(x-1)^3} = -1$

$\lim\limits_{h \to 0} \frac{h(6+\lambda\cosh) - \mu\sinh}{h^3} = -1$

$\lim\limits_{h \to 0} \frac{h\left(6+\lambda\left(1-\frac{h^2}{2}\right)\right) - \mu\left(h - \frac{h^3}{6}\right)}{h^3}$

$\lim\limits_{h \to 0} \frac{6h + \lambda h - \frac{\lambda h^3}{2} - \mu h + \frac{\mu h^3}{6}}{h^3}$

$\lim\limits_{h \to 0} \frac{h(6+\lambda-\mu) + h^3\left(\frac{\mu}{6} - \frac{\lambda}{2}\right)}{h^3}$

$\lambda - \mu + 6 = 0$ ...(1) & $\frac{\mu}{6} - \frac{\lambda}{2} = -1$ ...(2)

$\mu - \lambda = 6$ $\frac{\mu - 3\lambda}{6} = -1$

$\mu - 3\lambda = -6$ $\mu - 3\lambda = -6$

$2\lambda = 12$

$\lambda = 6 \qquad \mu = 12$

$\Rightarrow \lambda + \mu = 18$

93. $\lim\limits_{x\to 0^+} \dfrac{\tan\left(5(x)^{\frac{1}{3}}\right)\log_e\left(1+3x^2\right)}{\left(\tan^{-1}3\sqrt{x}\right)^2\left(e^{5(x)^{\frac{4}{3}}}-1\right)}$ is equal to :

$\lim\limits_{x\to 0^+} \dfrac{\tan\left(5(x)^{\frac{1}{3}}\right)\log_e\left(1+3x^2\right)}{\left(\tan^{-1}3\sqrt{x}\right)^2\left(e^{5(x)^{\frac{4}{3}}}-1\right)}$ बराबर है :

(A) $\dfrac{1}{15}$ (B) $\dfrac{5}{3}$ (C) 1 (D) $\dfrac{1}{3}$

Ans. D

Sol. $\lim\limits_{x\to 0^+}\left(\dfrac{\tan\left(5x^{\frac{1}{3}}\right)}{5x^{\frac{1}{3}}}\right)\left(\dfrac{\left(3\sqrt{x}\right)^2}{\left(\tan^{-1}3\sqrt{x}\right)^2}\right)\left(\dfrac{\ell\left(1+3x^2\right)}{3x^2}\right)\left(\dfrac{5x^{\frac{4}{3}}}{e^{5x^{\frac{4}{3}}}-1}\right)\times\dfrac{5x^{\frac{1}{3}}.3x^2}{5x^{\frac{4}{3}}.9x}$

$= \dfrac{1}{3}$

94. Given below are two statements :

Statement I : $\lim\limits_{x\to 0}\left(\dfrac{\tan^{-1}x+\log_e\sqrt{\dfrac{1+x}{1-x}}-2x}{x^5}\right)=\dfrac{2}{5}$

Statement II : $\lim\limits_{x\to 1}\left(x^{\frac{2}{1-x}}\right)=\dfrac{1}{e^2}$

In the light of the above statements, choose the ***correct*** answer from the options given below.

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

नीचे दो कथन दिए गए हैं :

कथन **I** : $\lim\limits_{x\to 0}\left(\dfrac{\tan^{-1}x+\log_e\sqrt{\dfrac{1+x}{1-x}}-2x}{x^5}\right)=\dfrac{2}{5}$

कथन **II** : $\lim\limits_{x\to 1}\left(x^{\frac{2}{1-x}}\right)=\dfrac{1}{e^2}$

इन कथनों के लिए नीचे दिए गए विकल्पों में से सही उत्तर चुनिए :

(A) कथन I सत्य है किंतु कथन II असत्य है

(B) कथन I असत्य है किंतु कथन II सत्य है

(C) कथन I और कथन II दोनों सत्य हैं

(D) कथन I और कथन II दोनों असत्य हैं

Ans. B

Sol. Statement-I : $\lim\limits_{x\to 0}\left(\dfrac{\tan^{-1}x+\log_e\sqrt{\dfrac{1+x}{1-x}}-2x}{x^5}\right)=\dfrac{2}{5}$

$$\Rightarrow \frac{\lim\limits_{x\to 0}\left(x-\frac{x^3}{3}+\frac{x^5}{5}+....\right)+\frac{1}{2}\left(\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right)+\frac{x^5}{5}.....\right)-\left(-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^5}{4}-\frac{x^5}{5}....\right)-2x}{x^5}$$

$$\lim_{x\to 0}\frac{\frac{2x^5}{5}}{x^5}+.....=\frac{2}{5}$$

Statement-II : $\lim\limits_{x\to 1}\left(x^{\frac{2}{1-x}}\right)=e^{\lim\limits_{x\to 1}(x-1)\cdot\frac{2}{1-x}}=e^{-2}=\dfrac{1}{e^2}$

95. If $\lim\limits_{x\to\infty}\left(\left(\dfrac{e}{1-e}\right)\left(\dfrac{1}{e}-\dfrac{x}{1+x}\right)\right)^x=\alpha$, then the value of $\dfrac{\log_e\alpha}{1+\log_e\alpha}$ equal to :

यदि $\lim\limits_{x\to\infty}\left(\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)\right)^x=\alpha$ है, तो $\frac{\log_e\alpha}{1+\log_e\alpha}$ का मान बराबर है :

(A) e (B) e^{-2} (C) e^2 (D) e^{-1}

Ans. A

Sol. $\lim\limits_{x\to\infty}\left(\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)\right)^x=\alpha$

$e^{\lim\limits_{x\to\infty}x\left\{\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)-1\right\}}=\alpha$

$e^{\lim\limits_{x\to\infty}x\left\{\frac{e(1+x-ex)}{(1-e)e(1+x)}-1\right\}}=\alpha$

$e^{\lim\limits_{x\to\infty}x\left\{\frac{e+ex-e^2x-e+e^2-ex+e^2x}{e+e^2+ex-e^2x}\right\}}=\alpha$

$e^{\lim\limits_{x\to\infty}\frac{xe^2}{e-e^2+x(e-e^2)}}=\alpha$

$e^{\frac{e^2}{e-e^2}}=\alpha \qquad \Rightarrow \alpha=e^{\frac{e}{1-e}}$

Now $\frac{\log\alpha}{1+\log_e\alpha}=\frac{\log_e\left(e^{\frac{e}{1-e}}\right)}{1+\log_e\left(e^{\frac{e}{1-e}}\right)}=\frac{\frac{e}{1-e}}{1+\frac{e}{1-e}} \Rightarrow e$

96. Let $f:R-\{0\}\to R$ be a function such that $f(x)-6f\left(\frac{1}{x}\right)=\frac{35}{3x}-\frac{5}{2}$. If the $\lim\limits_{x\to0}\left(\frac{1}{\alpha x}+f(x)\right)=\beta;\ \alpha,\beta\in R$, then $\alpha+2\beta$ is equal to :

माना कि $f:R-\{0\}\to R$ एक फलन इस प्रकार है कि $f(x)-6f\left(\frac{1}{x}\right)=\frac{35}{3x}-\frac{5}{2}$ है। यदि $\lim\limits_{x\to0}\left(\frac{1}{\alpha x}+f(x)\right)=\beta;\ \alpha,\beta\in$ R है, तो $\alpha+2\beta$ बराबर है :

(A) 3 (B) 5 (C) 4 (D) 6

Ans. C

Sol. $f(x)-6f\left(\frac{1}{x}\right)=\frac{35}{3x}-\frac{5}{2}$ ...(1)

Replace $x\to\frac{1}{x}$

$$f\left(\frac{1}{x}\right)-6f\left(x\right)=\frac{35x}{3}-\frac{5}{2} \qquad ...(2)$$

from (1) and (2)

$$f\left(x\right)=-2x-\frac{1}{3x}+\frac{1}{2}$$

Now,

$$\beta=\lim_{x\to0}\left(\frac{1}{\alpha x}-2x-\frac{1}{3x}+\frac{1}{2}\right)$$

$$\Rightarrow \alpha=3,\ \beta=\frac{1}{2}$$

So, $\alpha+2\beta=3+1=4$

97. For some a, b, let $f(x)=\begin{vmatrix} a+\frac{\sin x}{x} & 1 & b \\ a & 1+\frac{\sin x}{x} & b \\ a & 1 & b+\frac{\sin x}{x}\end{vmatrix}$, $x\neq0$, $\lim\limits_{x\to0}f(x)=\lambda+\mu a+\nu b$. Then $(\lambda+\mu+\nu)^2$ is equal to :

किसी a, b, के लिए माना कि $f(x)=\begin{vmatrix} a+\frac{\sin x}{x} & 1 & b \\ a & 1+\frac{\sin x}{x} & b \\ a & 1 & b+\frac{\sin x}{x}\end{vmatrix}$, $x\neq0$, $\lim\limits_{x\to0}f(x)=\lambda+\mu a+\nu b$ है। तब $(\lambda+\mu+\nu)^2$ बराबर है :

(A) 9 (B) 36 (C) 16 (D) 25

Ans. C

Sol. $\lim\limits_{x\to0}f\left(x\right)=\begin{vmatrix} a+1 & 1 & b \\ a & 2 & b \\ a & 1 & b+1\end{vmatrix}$

$= (a + 1)(b + 2) - a(1) + a(-b)$
$= 2 + a + b$
$\lambda = 2,\ \mu = 1,\ \nu = 1$
$(\lambda + \mu + \nu)^2 = 16$

98. $\lim\limits_{n\to\infty}\dfrac{\left(1^2-1\right)(n-1)+\left(2^2-2\right)(n-2)+\cdots+\left((n-1)^2-(n-1)\right)\cdot 1}{\left(1^3+2^3+\cdots+n^3\right)-\left(1^2+2^2+\cdots+n^2\right)}$ is equal to :

$\lim\limits_{n\to\infty}\dfrac{\left(1^2-1\right)(n-1)+\left(2^2-2\right)(n-2)+\cdots+\left((n-1)^2-(n-1)\right)\cdot 1}{\left(1^3+2^3+\cdots+n^3\right)-\left(1^2+2^2+\cdots+n^2\right)}$ बराबर है :

(A) $\dfrac{2}{3}$ (B) $\dfrac{3}{4}$ (C) $\dfrac{1}{2}$ (D) $\dfrac{1}{3}$

Ans. D

Sol. $\lim\limits_{n\to\infty}\dfrac{\left(1^2-1\right)(n-1)+\left(2^2-2\right)(n-2)+\ldots+\left(\left(n-1\right)^2-\left(n-1\right)\right)\times 1}{\left(1^3+2^3+\ldots..+n^3\right)\left(1^2+2^2+\ldots.+n^2\right)}$

Numerator $=\sum\limits_{r=1}^{n-1}\left(\left(r-1\right)^2-\left(r-1\right)\right)\left(n-r\right)$

$=\sum\limits_{r=1}^{n-1}\left(r-1\right)-\left(r-2\right)\left(n-r\right)$

$=\sum\limits_{r=1}^{n-1}-r^3+\left(n+3\right)r^2-\left(2+3n\right)r+2n$

We will take term with the greatest power of n

$=\dfrac{-1}{4}n^4+\dfrac{1}{3}+\dfrac{1}{3}n^4=\dfrac{1}{12}n^4$

Denominator $=\sum\limits_{r=1}^{n}r^3-\sum\limits_{r=1}^{n}r^2$

$=\left(\dfrac{n\left(n+1\right)^2}{2}\right)-\left(\dfrac{n\left(n+1\right)\left(2n+1\right)}{6}\right)$

Greatest power of n is $\dfrac{n^4}{4}$

$\lim\limits_{n\to\infty}\dfrac{\frac{1}{12}n^4}{\frac{n^4}{4}}=\dfrac{1}{3}$

99. Let a circle passing through (2, 0) have its centre at the point (h, k). Let (x_c, y_c) be the point of intersection of the

lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$. If $h = \lim_{c\to 1} x_c$ and $k = \lim_{c\to 1} y_c$, then the equation of the circle is:

माना (2, 0) से होकर जाने वाले एक वृत्त का केन्द्र (h, k) है। माना रेखाओं $3x + 5y = 1$ तथा $(2 + c)x + 5c^2y = 1$ का प्रतिच्छेदन बिन्दु (x_c, y_c) है। यदि $h = \lim_{c\to 1} x_c$ तथा $k = \lim_{c\to 1} y_c$ है, तो वृत्त का समीकरण है :

(A) $25x^2 + 25y^2 - 2x + 2y - 60 = 0$ (B) $5x^2 + 5y^2 - 4x + 2y - 12 = 0$

(C) $25x^2 + 25y^2 - 20x + 2y - 60 = 0$ (D) $5x^2 + 5y^2 - 4x - 2y - 12 = 0$

Ans. C

Sol. $3x + 5y = 1$

$(2 + c)x + 2c^2y = 1$

$(3c^2 - c - 2)x = (c^2 - 1)$

$$x = \frac{c^2-1}{3c^2-c-2} \text{ and } y = \frac{1}{5}\left(1 - \frac{3(c^2-1)}{3c^2-c-2}\right) = \frac{1-c}{5(3c^2-c-2)}$$

$$h = \lim_{c\to 1} \frac{(1-c)}{(1-c)}\left(\frac{1+c}{2+3c}\right) = \frac{2}{5}$$

$$k = \lim_{c\to 1} \frac{1-c}{5(c-1)(3c+2)} = \lim_{c\to 1} \frac{-1}{5(3c+12)} = -\frac{1}{25}$$

$$\text{Centre of circle} = \left(\frac{2}{5}, -\frac{1}{25}\right)$$

$$\text{Now radius of circle } \sqrt{\left(2-\frac{2}{5}\right)^2 + \left(\frac{1}{25}\right)^2} = \sqrt{\frac{64}{25} + \frac{1}{625}}$$

Equation of circle is

$$\left(x - \frac{2}{5}\right)^2 + \left(y + \frac{1}{25}\right)^2 = \frac{64}{25} + \frac{1}{625}$$

$$x^2 + y^2 - \frac{4}{5}x + \frac{2}{25}y + \frac{4}{25} + \frac{1}{625} = \frac{64}{25} + \frac{1}{625}$$

$25x^2 + 25y^2 - 20x + 2y + 4 = 64$

$25x^2 + 25y^2 - 20x + 2y - 60 = 0$

100. $\lim\limits_{x\to 0}\dfrac{e-(1+2x)^{\frac{1}{2x}}}{x}$ is equal to :

$\lim\limits_{x\to 0}\dfrac{e-(1+2x)^{\frac{1}{2x}}}{x}$ बराबर है :

(A) e (B) $\dfrac{-2}{e}$ (C) 0 (D) $e - e^2$

Ans. A

Sol. $\text{Lim}_{x\to 0}\dfrac{e-e^{\frac{1}{2x}\ell n(1+2x)}}{x}$

$= \text{Lim}_{x\to 0}(-e)\dfrac{\left(e^{\frac{\ell n(1+2x)}{2x}-1}-1\right)}{x}$

$= \text{Lim}_{x\to 0}(-e)\dfrac{\ell n(1+2x)-2x}{2x^2}$

$= (-e)\times(-1)\dfrac{4}{2\times 2} = e$

101. If $a = \lim\limits_{x\to 0}\dfrac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ and $b = \lim\limits_{x\to 0}\dfrac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$, then the value of ab^3 is :

यदि If $a = \lim\limits_{x\to 0}\dfrac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ तथा $b = \lim\limits_{x\to 0}\dfrac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$ हैं, तो ab^3 का मान है :

(A) 36 (B) 25 (C) 32 (D) 30

Ans. C

Sol. $a = \lim\limits_{x\to 0}\dfrac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$

$= \lim\limits_{x\to 0}\dfrac{\sqrt{1+x^4}-1}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)}$

$= \lim\limits_{x\to 0}\dfrac{x^4}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)\left(\sqrt{1+x^4}+\right.}$

Applying limit $a = \frac{1}{4\sqrt{2}}$

$$b = \lim_{x\to 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1+\cos x}}$$

$$= \lim_{x\to 0} \frac{(1-\cos x)(\sqrt{2} + \sqrt{1+\cos x})}{2-(1+\cos x)}$$

$$b = \lim_{x\to 0} (1+\cos x)(\sqrt{2} + \sqrt{1+\cos x})$$

Applying limit $b = 2(\sqrt{2} + \sqrt{2}) = 4\sqrt{2}$

Now, $ab^3 = \frac{1}{4\sqrt{2}} \times (4\sqrt{2})^3 = 32$

102. $\lim_{x\to 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2} =$

(A) does not exist (B) is equal to 1

(C) is equal to 2 (D) is equal to –1

$\lim_{x\to 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$:

(A) का अस्तित्व नहीं है (B) 1 के बराबर है

(C) 2 के बराबर है (D) –1 के बराबर है

Ans. C

Sol. $\lim_{x\to 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$

Since, it is an even function

$\Rightarrow$ LHL = RHL

$$RHL = \lim_{x\to 0^+} \frac{e^{2\sin x} - 2\sin x - 1}{x^2}$$

$$= \lim_{x\to 0^+} \frac{e^{2\sin x}(2\cos x) - 2\cos x}{2x}$$

$$= \lim_{x\to 0^+} = \left(\frac{e^{2\sin x} - 1}{2\sin x}\right)\left(\frac{2\sin x}{x}\right)\cos x$$

$= 1 \times 2 \times 1$

$= 2$

103. Let $f :\to R \to (0,\infty)$ be strictly increasing function such that $\lim\limits_{x\to\infty}\dfrac{f(7x)}{f(x)}=1$. Then, the value of of $\lim\limits_{x\to\infty}\left[\dfrac{f(5x)}{f(x)}-1\right]$ is equal to :

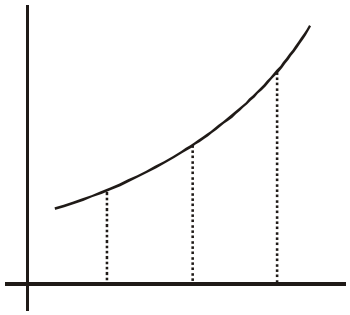
माना $f :\to R \to (0,\infty)$ निरंतर वर्धमान फलन है तथा $\lim\limits_{x\to\infty}\dfrac{f(7x)}{f(x)}=1$ है। तो $\lim\limits_{x\to\infty}\left[\dfrac{f(5x)}{f(x)}-1\right]$ का मान बराबर है :

(A) $\dfrac{7}{5}$ (B) 0 (C) 1 (D) 4

Ans. B

Sol. $f(x)\le f(5x)\le f(7x),\forall x>0$

$x<5x<7x$



$$1\le\frac{f(5x)}{f(x)}\le\frac{f(7x)}{f(x)}$$

$$\text{As } x\to\infty,\frac{f(5x)}{f(x)}\to 1^{+}$$

$$\lim_{x\to\infty}\left[\frac{f(5x)}{f(x)}-1\right]=0$$

104. Let $f(x)=\begin{cases}x-1, x \text{ is even},\\ 2x, \quad x \text{ is odd},\end{cases}$ $x\in N$. If for some $a\in N, f\left(f\left(f(a)\right)\right)=21$, then $\lim\limits_{x\to a^{-}}\left\{\dfrac{|x|^3}{a}-\left[\dfrac{x}{a}\right]\right\}$, where [t] denotes the greatest integer less than or equal to t, is equal to :

माना $f(x)=\begin{cases}x-1, x \text{ is even},\\ 2x, \quad x \text{ is odd},\end{cases}$ $x\in N$ यदि किसी $a\in N$ के लिए $f\left(f\left(f(a)\right)\right)=21$, तो $\lim\limits_{x\to a^{-}}\left\{\dfrac{|x|^3}{a}-\left[\dfrac{x}{a}\right]\right\}$, जहाँ [t] के सम्मान या कम महत्तम पूर्णांक दर्शाता है, बराबर होगा :

(A) 225 (B) 144 (C) 121 (D) 169

Ans. B

Sol. $f(\underline{f(f(a))}) = 21$

$f(f(a)) - 1 = 21$

$f(f(a)) = 22$

$2f(a) = 22$

$f(a) = 11$

$a - 1 = 11$

$\Rightarrow a = 12$

$$\lim_{x\to 12}\left(\frac{|x|^3}{12} - \left[\frac{x}{12}\right]\right)$$

$= 144 - 0 = 144$

105. Let $a_1, a_2, a_3, \ldots, a_n$ be n positive consecutive terms of an arithmetic progression. if $d > 0$ is its common difference, then $\lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \ldots\ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$ is :

माना $a_1, a_2, a_3, \ldots, a_n$ समांतर श्रेणी के धनात्मक क्रमागत n पद हैं। यदि सार्वन्तर $d > 0$ है, तब

$\lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \ldots\ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$ का मान है :

(A) 1 (B) 0 (C) $\sqrt{d}$ (D) $\frac{1}{\sqrt{d}}$

Ans. A

Sol. $\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + - - - + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$

$$= \left(\frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1} + \frac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2} + \frac{\sqrt{a_4}-\sqrt{a_3}}{a_4-a_3} + - - - + \frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{a_n-a_{n-1}}\right)$$

$$= \frac{1}{d}\left(\sqrt{a_n} - \sqrt{a_1}\right)$$

$$\therefore \lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + - - - + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$$

$$= \lim_{n\to\infty}\sqrt{\frac{d}{n}}\left(\frac{\sqrt{a_n}-\sqrt{a_1}}{d}\right)$$

$$= \lim_{n\to\infty}\left(\frac{\sqrt{a_1+(n-1)d}-\sqrt{a_1}}{\sqrt{n}\sqrt{d}}\right) = \lim_{n\to\infty}\left(\sqrt{\frac{a_1}{nd}+\left(1-\frac{1}{n}\right)}-\sqrt{\frac{a_1}{nd}}\right) = 1$$

106. $\lim\limits_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$ is equal to :

$\lim\limits_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$ बराबर है :

(A) 1 (B) $\sqrt{2}$ (C) 0 (D) $\frac{1}{\sqrt{2}}$

Ans. C

Sol. $P = \lim\limits_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)$

Let

$2^{\frac{1}{2}}-2^{\frac{1}{3}} \rightarrow$ Smallest

$2^{\frac{1}{2}}-2^{\frac{1}{2n+1}} \rightarrow$ Largest

Sandwich th.

$$\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n \le P \le \left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n$$

$\begin{pmatrix}\text{lie b/w}\\ 0 \text{ and } 1\end{pmatrix}^n$

$$\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n = 0$$

$$\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n = 0$$

$\therefore P = 0$

107. For $t > -1$, let α_t and β_t be the roots of the equation $\left((t+2)^{\frac{1}{7}}-1\right)x^2+\left((t+2)^{\frac{1}{6}}-1\right)x+\left((t+2)^{\frac{1}{21}}-1\right)=0$. If

$\lim\limits_{t\to -1^+}\alpha_t = a$ and $\lim\limits_{t\to -1^+}\beta_t = b$, then $72(a+b)^2$ is equal to _____.

माना $t > -1$ के लिए समीकरण $\left((t+2)^{\frac{1}{7}}-1\right)x^2+\left((t+2)^{\frac{1}{6}}-1\right)x+\left((t+2)^{\frac{1}{21}}-1\right)=0$ के मूल α_t और β_t हैं। यदि $\lim\limits_{t\to -1^+}\alpha_t = a$ और $\lim\limits_{t\to -1^+}\beta_t = b$ हैं, तो $72(a+b)^2$ बराबर है _____.

Ans. 98

Sol. $\alpha_t+\beta_t=\dfrac{(t+2)^{\frac{1}{6}}-1}{(t+2)^{\frac{1}{7}}-1}$

$$\lim_{t\to -1^+}(\alpha_t+\beta_t)=\frac{\frac{1}{6}}{\frac{1}{7}}=\frac{7}{6}$$

$$\Rightarrow \lim_{t\to -1^+}\alpha_t+\lim_{t\to -1^+}\beta_t=\frac{7}{6}$$

S0, $a+b=\dfrac{7}{6}$

$$72(a+b)^2=72\cdot\frac{49}{36}=98$$

108. Let $f(x)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\left(\dfrac{\tan\left(\frac{x}{2^{r+1}}\right)+\tan^3\left(\frac{x}{2^{r+1}}\right)}{1-\tan^2\left(\frac{x}{2^{r+1}}\right)}\right)$. Then $\lim\limits_{x\to 0}\dfrac{e^x-e^{f(x)}}{(x-f(x))}$ is equal to _____.

माना $f(x)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\left(\dfrac{\tan\left(\frac{x}{2^{r+1}}\right)+\tan^3\left(\frac{x}{2^{r+1}}\right)}{1-\tan^2\left(\frac{x}{2^{r+1}}\right)}\right)$ है। तब $\lim\limits_{x\to 0}\dfrac{e^x-e^{f(x)}}{(x-f(x))}$ बराबर है _____।

Ans. 1

Sol. $f(x)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\left(\dfrac{\tan\theta+\tan^3\theta}{1-\tan^2\theta}\right)$

$\theta=\left(\dfrac{x}{2^{r+1}}\right)$

$$f(x)=\sum_{r=1}^{n}(\tan\theta)\frac{(1+\tan^2\theta)}{(1-\tan^2\theta)}$$

$$f(x)=\sum_{r=1}^{n}\frac{(\tan\theta)}{(\cos 2\theta)}$$

$$f(x)=\sum_{r=1}^{n}\frac{\sin(2\theta-\theta)}{\cos\theta\cos 2\theta}$$

$$f(x)=\sum_{r=1}^{n}\left(\frac{\sin 2\theta\cdot\cos\theta-\cos 2\theta\sin\theta}{\cos\theta\cos 2\theta}\right)$$

$$f(x)=\sum_{r=1}^{n}(\tan 2\theta-\tan\theta)$$

$$f(x)=\sum_{r=1}^{n}\left[\tan\left(\frac{x}{2^r}\right)-\tan\left(\frac{x}{2^{r+1}}\right)\right]$$

$$f(x)=\tan\left(\frac{x}{2}\right)-\tan\left(\frac{x}{4}\right)$$

$$+$$

$$\tan\left(\frac{x}{4}\right)-\tan\left(\frac{x}{8}\right)$$

$$+$$

$$\vdots$$

$$\tan\left(\frac{x}{2^n}\right)-\tan\left(\frac{x}{2^{n+1}}\right)$$

$$f(x)=\lim_{n\to\infty}\tan\left(\frac{x}{2}\right)-\tan\left(\frac{x}{2^{n+1}}\right)$$

$$f(x)=\tan\left(\frac{x}{2}\right)$$

$$=\lim_{x\to 0}\frac{e^x-e^{\tan\left(\frac{x}{2}\right)}}{x-\tan\left(\frac{x}{2}\right)}$$

By series expansion of $e^s = 1 + s$

$$\lim_{x\to 0}\frac{(1+x)-\left(1+\tan\left(\frac{x}{2}\right)\right)}{\left(x-\tan\left(\frac{x}{2}\right)\right)}=1$$

109. Let [t] be the greatest integer less than or equal to t. Then least value of $p \in N$ for which $\lim_{x\to 0^+}\left(x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+.....+\left[\frac{p}{x}\right]\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+...+\left[\frac{9^2}{x^2}\right]\right)\right)\geq 1$ is equal to _____.

माना कि [t], t से छोटा या उसके बराबर का महत्तम पूर्णांक है। तब $p \in N$ का न्यूनतम मान, जिसके लिए $\lim_{x\to 0^+}\left(x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+.....+\left[\frac{p}{x}\right]\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+...+\left[\frac{9^2}{x^2}\right]\right)\right)\geq 1$ है, बराबर है _____.

Ans. 24

Sol. $\lim_{x\to 0^+}\left(x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+.....+\left[\frac{P}{x}\right]\right)\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+....\left[\frac{9^2}{x^2}\right]\right)\geq 1$

$x-1<[x]\leq x$

$\frac{1}{x}-1<\left[\frac{1}{x}\right]\leq\frac{1}{x}$

$\frac{P(P+1)}{2x}-P<.....\leq\frac{P(P+1)}{2x}$

$\frac{P(P+1)}{2}-Px<....\leq\frac{P(P+1)}{2}$

$P^2+P-572\geq 0$

$P=\frac{-1\pm\sqrt{1+4\times 572}}{2}$

$P=\frac{-1\pm\sqrt{2289}}{2}$

$P=\frac{46.84}{2}$ or $-\frac{48.84}{2}$

$P = 23.42$ or -24.42

$(P + 24.42)(P - 23.42)\geq 0$

$P\in(-\infty,-24.42]\cup[23.42,+\infty)$

$P = 24$

110. If $\lim\limits_{x\to 1}\dfrac{(5x+1)^{\frac{1}{3}}-(x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}}-(x+4)^{\frac{1}{2}}}=\dfrac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$, where gcd(m, n) = 1, then 8m + 12n is equal to _____.

यदि $\lim\limits_{x\to 1}\dfrac{(5x+1)^{\frac{1}{3}}-(x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}}-(x+4)^{\frac{1}{2}}}=\dfrac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$, जहाँ gcd(m, n) = 1 हैं, तो 8m + 12n बराबर है _____

Ans. 100

Sol. $\ell=\lim\limits_{x\to 1}\dfrac{(5x+1)^{\frac{1}{3}}-(x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}}-(x+4)^{\frac{1}{2}}}$

From : $\dfrac{0}{0}$, using L-H rule

$$\ell=\lim_{x\to 1}\frac{\frac{1}{3}\times 5(5x+1)^{-\frac{2}{3}}-\frac{1}{3}(x+5)^{-\frac{2}{3}}}{\frac{1}{2}\times 2(2x+3)^{-\frac{1}{2}}-\frac{1}{2}(x+4)^{-\frac{1}{2}}}$$

$$=\frac{\left(\frac{5}{3}-\frac{1}{3}\right)6^{-\frac{2}{3}}}{\frac{1}{2}5^{-\frac{1}{2}}}=\frac{8}{3}\times\frac{5^{\frac{1}{2}}}{6^{\frac{2}{3}}}=\frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$$

$\Rightarrow m = 8, n = 3$

$\Rightarrow 8m + 12n = 100$

111. Let a > 0 be a root of the equation $2x^2 + x - 2 = 0$. If $\lim\limits_{x\to\frac{1}{a}}\dfrac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{(1-ax)^2}=\alpha+\beta\sqrt{17}$ where $\alpha, \beta \in Z$, then $\alpha + \beta$ is equal to _____.

माना समीकरण $2x^2 + x - 2 = 0$ का एक मूल $a > 0$ है। यदि $\lim\limits_{x\to\frac{1}{a}}\dfrac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{(1-ax)^2}=\alpha+\beta\sqrt{17}$ है, जहाँ $\alpha, \beta \in Z$ है। तो $\alpha + \beta$ बराबर है _____

Ans. 170

Sol. $\therefore 2x^2 + x - 2 = 0$ has two roots where

$a = \frac{\sqrt{17}-1}{4}$ and another root is $\frac{-\sqrt{17}-1}{4}$

And $2 + x - 2x^2 = -2\left(x - \frac{1}{a}\right)\left(x + \frac{4}{\sqrt{17}+1}\right)$

Now $\lim\limits_{x\to\frac{1}{a}} \frac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{(1-ax)^2}$

$\lim\limits_{x\to\frac{1}{a}} \frac{32\sin^2\left(2+x-2x^2\right)}{a^2\left(\frac{1}{a}-x\right)^2}$

$= \lim\limits_{x\to\frac{1}{a}} \frac{\left(x+\frac{4}{\sqrt{17}+1}\right)^2 32\cdot\left(\sin\left(\frac{1}{2}\cdot(-2)\right)\left(x-\frac{1}{a}\right)\left(x+\frac{4}{\sqrt{17}+1}\right)\right)^2}{a^2\left(\left(x-\frac{1}{a}\right)\left(x+\frac{4}{\sqrt{17}+1}\right)\right)^2}$

$= 2\cdot\left(\frac{1}{a}+\frac{4}{\sqrt{17}+1}\right)^2\cdot\left(\frac{4}{\sqrt{17}-1}\right)^2$

$= 2\left(\frac{4}{\sqrt{17}-1}+\frac{4}{\sqrt{17}+1}\right)^2\cdot\left(\frac{4}{\sqrt{17}-1}\right)^2$

$= \frac{17\times 4}{18-2\sqrt{17}} = \frac{68}{9-\sqrt{17}}$

$= 17\left(9+\sqrt{17}\right)$

$\alpha + \beta = 170$

112. If $\alpha = \lim\limits_{x\to 0^+}\left(\frac{e^{\sqrt{\tan x}}-e^{\sqrt{x}}}{\sqrt{\tan x}-\sqrt{x}}\right)$ and $\beta = \lim\limits_{x\to 0}(1+\sin)^{\frac{1}{2}\cot x}$ are the roots of the quadratic equation $ax^2 + bx - \sqrt{e} = 0$, then $12\log_e(a+b)$ is equal to _____.

यदि $\alpha = \lim\limits_{x\to 0^+}\left(\frac{e^{\sqrt{\tan x}}-e^{\sqrt{x}}}{\sqrt{\tan x}-\sqrt{x}}\right)$ तथा $\beta = \lim\limits_{x\to 0}(1+\sin)^{\frac{1}{2}\cot x}$ द्विघात समीकरण $ax^2 + bx - \sqrt{e} = 0$ के मूल हैं, तो

$12 \log_e(a + b)$ बराबर है _____.

Ans. 6

Sol. $\alpha = \lim\limits_{x\to 0^+} \dfrac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\left(\sqrt{\tan x} - \sqrt{x}\right)}$

$$= \lim_{x\to 0} \frac{e^{\sqrt{x}}\left(e^{\sqrt{\tan x}-\sqrt{x}} - 1\right)}{\left(\sqrt{\tan x} - \sqrt{x}\right)} = 1$$

$$\beta = \lim_{x\to 0}(1+\sin x)^{\frac{1}{2}\cot x} = \lim_{x\to 0} e^{(\sin x)\left(\frac{1}{2}\cot x\right)}$$

$$= \lim_{x\to 0} e^{\frac{1}{2}\cos x} = e^{\frac{1}{2}}$$

$$\text{Product of roots} = \sqrt{e} = \frac{-\sqrt{e}}{a} \Rightarrow a = -1$$

$$\text{Sum of roots} = \frac{-b}{a} = 1 + \sqrt{e}$$

$$= b = \sqrt{e} + 1$$

$$\Rightarrow 12\ell n(a+b) = 12\ell n\left(\sqrt{e} + 1 - 1\right) = 12\ell n\left(e^{\frac{1}{2}}\right) = 6$$